INCH-POUND
MIL-DTL-24643/8F
27 November 2023
SUPERSEDING
MIL-DTL-24643/8E
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TYPE LSMMOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First	-	Copper conductor, coated or uncoated, 24 AWG, class O stranding.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Minimum average wall thickness of 0.020 inch. Standard identification code applied by method 1 or method 3. If identification method 1 is used, the insulation color shall be black.
Fourth	-	Five conductors shall be cabled together with a left-hand lay as required for the completed cable to meet the twisting and bending endurance tests. A tie cord shall be laid straight as a central core. Fillers may be used as necessary to form a firm, well-rounded assembly. Cabling sequence shall be consecutive.
Fifth	-	An optional binder.
Sixth	-	Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/8	Type and size	Number of conductors	Insulated conductor diameter min. (inch)	Overall diameter max. (inch)
-01UN	LSMMOP-5	5	0.065	0.305

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	27.96
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	1,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	100
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Twisting and bending endurance	
At 167 °F (75 °C) (cycles, min.)	6,000
At -22±3.6 °F (-30±2 °C) (cycles, min.)	2,000
Roller diameter (inches, max.)	1.5
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

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Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation, method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45

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Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSMMOP-5	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-024)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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